Python Documentation

This documentation will serve as a guide to the students of Yverdon de Pestalozzi school inc. on what they will learn, use and code for the python programming language. The source of this documentation comes from the official website of python. *https://docs.python.org/release/3.11.0/tutorial/index.html*

This documentation will also tackle about the difficulty variants inside the game. There would be 3 difficulties including:

Easy – Which will talk about the basics of python.

Medium - Which will talk about the more important parts of python.

Difficult or Hard – Which will talk about a combination of easy and medium difficulty to code for an algorithmica bots inside the game. Plus, more parts of python.

Python is an easy to learn, powerful programming language. It has efficient high-level data structures and a simple but effective approach to object-oriented programming. Python also has a very readable syntax in which students can comprehend it easily than other programming languages.

Python 3.11.0. was released on October 24, 2022. The reason in which we will not use the current version of python is that the newer added features is not necessary to study since it is just the adjustments of the past features. We will focus on the fundamentals of python.

These are the basics of python and the difficulties of the game.

Easy: The students will learn the python basics in this category.

Print – The students will be able to print a word. ex. print("Hello world")

Int - The students shall learn how to declare an integer or any whole number. x = 10

Float - The students shall learn how to declare a float or any number with decimal.

Ex. x = 10.5

Boolean - The students shall learn how does Booleans work. Ex. Is_rainy = True

Type casting – The students shall learn how to type cast from int to float etc.

Ex. gpa = 1.5

gpa = int(gpa)

User Input – The students shall learn how to get the user input.

Ex. input("Enter your name: ”)

User input with int and float – The students shall learn how to get the user input with numbers. Ex. int(input(“enter num. ”)) float(input(“enter num. ”)).

Concatenation – The students shall learn how to concatenate

ex. age = 18

print(f”hi your age is {age}”)

Medium: The students will learn a little more advanced in python.

If else statements – The students shall learn how to use if and else statements.

Ex. if age > 18:

print(“You can vote now”)

else:

print("You are too young to vote”)

Logical Operators – The students shall learn how to use logical operators.

Ex. is_rainy AND is_cloudy, is_rainy OR is_cloudy, is_rainy not is_cloudy.

Import math and its functions – The students shall learn how to import math in their code and use its functions such as: pow(), round(), ceil(), floor(), and sqrt()

String methods – The students shall learn how to use the string method such as:

Len(), find(), rfind(), capitalize(), upper(), lower(), isalpha(), count(), and replace()

Array – The students shall learn how to declare an array/linked list

Ex. variable = []

Hard: The students will learn additional advanced codes in python.

Looping – The students shall learn how to code while loops in python.

Ex. number = 0

while True:

print(f"Number is {number}!")

number += 1

if number >= 10:

break

Function – The students shall learn how to use, call and code a function code.

Ex. def my_function():
print("Hello from a function")

Random numbers – The students shall learn how to Import random numbers.

Ex. seed(), getstate(), setstate(), randrange() randit()

Exception Handling – The students shall learn how exception handling works.

Ex. while True:

try:

x = int(input("Please enter a number: "))

break

except ValueError:

print("Oops! That was no valid number. Try again...")

Iterators – The students shall learn how to use iterators.

Ex. for element **in** [1, 2, 3]:

  print(element)

for element **in** (1, 2, 3):

  print(element)

for key **in** {'one':1, 'two':2}:

  print(key)

for char **in** "123":

  print(char)

for line **in** open("myfile.txt"):

  print(line, end='')

Class – the students shall learn how to code for class in python.

Ex. class ClassName:

  <statement-1>

  .

  .

  .

  <statement-N>

Inheritance – The students shall learn how to use and call inheritance.

Ex. class DerivedClassName(BaseClassName):

  <statement-1>

  .

  .

  .

  <statement-N>

Dates and times – The students will learn how to call and import the date and time.

Ex. **from datetime import** date

These are the available syntaxes and codes that the students shall learn when playing the game. This codes will give them an insight on how does programming languages work in their way. The researchers will add more features about python in the future of the development.